Importing Data

In [4]:
```
%load_ext sql
```

```
The sql extension is already loaded. To reload it, use:
  %reload_ext sql
```

In [2]:
```
%sql mssql+pyodbc://DESKTOP-TANM3JG/Amazon?driver=ODBC+Driver+17+for+SQL+Server&tru
```

In [3]:
```
%%sql
SELECT TOP 5 *
FROM [Sample - Superstore];
```

```
 * mssql+pyodbc://DESKTOP-TANM3JG/Amazon?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
Done.
```

Out[3]:

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Seg
1	CA-2016-152156	2016-11-08 00:00:00	2016-11-11 00:00:00	Second Class	CG-12520	Claire Gute	Con:
2	CA-2016-152156	2016-11-08 00:00:00	2016-11-11 00:00:00	Second Class	CG-12520	Claire Gute	Con:
3	CA-2016-138688	2016-06-12 00:00:00	2016-06-16 00:00:00	Second Class	DV-13045	Darrin Van Huff	Corp
4	US-2015-108966	2015-10-11 00:00:00	2015-10-18 00:00:00	Standard Class	SO-20335	Sean O'Donnell	Con:
5	US-2015-108966	2015-10-11 00:00:00	2015-10-18 00:00:00	Standard Class	SO-20335	Sean O'Donnell	Con:

EDA

1-what are the sales by city?

2-Customers by Transactions?

3-customer segmentation

4-sales over Time?

5-sales by segment?

sales by city

In [9]:
```
%%sql
SELECT TOP(10) City, round(SUM(Sales),2)AS TotalSales
FROM [Sample - Superstore]
GROUP BY City
ORDER BY TotalSales DESC;
```

```
 * mssql+pyodbc://DESKTOP-TANM3JG/Amazon?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
Done.
```

Out[9]:

City	TotalSales
New York City	256368.16
Los Angeles	175851.34
Seattle	119540.74
San Francisco	112669.09
Philadelphia	109077.01
Houston	64504.76
Chicago	48539.54
San Diego	47521.03
Jacksonville	44713.18
Springfield	43054.34

Customer by Transation

In [13]:
```
%%sql
select Top(10) Customer_Name , Count(Order_ID) as Transation_NUM
from [Sample - Superstore]
group by Customer_Name
order by Transation_NUM desc
```

```
 * mssql+pyodbc://DESKTOP-TANM3JG/Amazon?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
Done.
```

Out[13]:

Customer_Name	Transation_NUM
William Brown	37
John Lee	34
Matt Abelman	34
Paul Prost	34
Seth Vernon	32
Edward Hooks	32
Chloris Kastensmidt	32
Jonathan Doherty	32
Arthur Prichep	31
Zuschuss Carroll	31

Customer Segmentation

In [14]:

```
%%sql
Create view Customers_Levels AS
select Customer_Name ,round(SUM(Sales),1) as TotaL_Sales,
Case
    when SUM(Sales)>=10000 then 'VIP Customer'
    when SUM(Sales)>=5000 then 'AVG Customer'
    else  'Normal Customer'
end as Customer_level

from [Sample - Superstore]
group by Customer_Name;
```

```
 * mssql+pyodbc://DESKTOP-TANM3JG/Amazon?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
(pyodbc.ProgrammingError) ('42S01', "[42S01] [Microsoft][ODBC Driver 17 for SQL Serv
er][SQL Server]There is already an object named 'Customers_Levels' in the database.
(2714) (SQLExecDirectW)")
[SQL: Create view Customers_Levels AS
select Customer_Name ,round(SUM(Sales),1) as TotaL_Sales,
Case
    when SUM(Sales)>=10000 then 'VIP Customer'
    when SUM(Sales)>=5000 then 'AVG Customer'
    else  'Normal Customer'
end as Customer_level

from [Sample - Superstore]
group by Customer_Name;]
(Background on this error at: https://sqlalche.me/e/20/f405)
```

In [21]:

```
%%sql
select tOP(15)* from Customers_Levels;
```

```
 * mssql+pyodbc://DESKTOP-TANM3JG/Amazon?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
Done.
```

Out[21]:

Customer_Name	TotaL_Sales	Customer_level
Michael Chen	3805.7	Normal Customer
Brian Moss	7294.2	AVG Customer
Tamara Chand	19052.2	VIP Customer
Justin MacKendrick	2833.9	Normal Customer
Doug Bickford	1989.0	Normal Customer
Jesus Ocampo	1090.8	Normal Customer
Sheri Gordon	1884.8	Normal Customer
John Huston	528.9	Normal Customer
Sean Miller	25043.0	VIP Customer
Liz Carlisle	2095.1	Normal Customer
Benjamin Patterson	1181.5	Normal Customer
Shahid Hopkins	2180.7	Normal Customer
Tracy Collins	742.6	Normal Customer
Eugene Moren	4588.4	Normal Customer
Alice McCarthy	814.0	Normal Customer

In [16]:

```
%%sql
Select Customer_Level, count(Customer_Level) as customer_Count
from Customers_Levels
group by Customer_Level
order by customer_Count desc;
```

```
 * mssql+pyodbc://DESKTOP-TANM3JG/Amazon?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
Done.
```

Out[16]:

Customer_Level	customer_Count
Normal Customer	676
AVG Customer	97
VIP Customer	20

In [17]:

```
%%sql
select * from Customers_Levels
where Customer_Level ='VIP Customer';
```

```
 * mssql+pyodbc://DESKTOP-TANM3JG/Amazon?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
Done.
```

Out[17]:

Customer_Name	TotaL_Sales	Customer_level
Tamara Chand	19052.2	VIP Customer
Sean Miller	25043.0	VIP Customer
Sanjit Chand	14142.3	VIP Customer
Hunter Lopez	12873.3	VIP Customer
Tom Ashbrook	14595.6	VIP Customer
Seth Vernon	11470.9	VIP Customer
Maria Etezadi	10663.7	VIP Customer
Edward Hooks	10310.9	VIP Customer
Greg Tran	11820.1	VIP Customer
Karen Ferguson	10604.3	VIP Customer
Raymond Buch	15117.3	VIP Customer
Christopher Conant	12129.1	VIP Customer
Clay Ludtke	10880.5	VIP Customer
Caroline Jumper	11165.0	VIP Customer
Becky Martin	11789.6	VIP Customer
Adrian Barton	14473.6	VIP Customer
Bill Shonely	10501.7	VIP Customer
Sanjit Engle	12209.4	VIP Customer
Ken Lonsdale	14175.2	VIP Customer
Todd Sumrall	11891.8	VIP Customer

sales over Time

by year

In [19]:

```
%%sql
select Year(Order_Date) as year ,
round(SUM(Sales),1) as TotaL_Sales
From [Sample - Superstore]
Group by Year(Order_Date)
order by year;
```

```
 * mssql+pyodbc://DESKTOP-TANM3JG/Amazon?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
Done.
```

Out[19]:

year	TotaL_Sales
2014	484247.5
2015	470532.5
2016	609205.6
2017	733215.3

By Month and year

In [20]:

```
%%sql
select Year(Order_Date) as year ,Month(Order_Date) as Month,
round(SUM(Sales),1) as TotaL_Sales
From [Sample - Superstore]
Group by Year(Order_Date),Month(Order_Date)
order by year, Month;
```

```
 * mssql+pyodbc://DESKTOP-TANM3JG/Amazon?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
Done.
```

Out[20]:

year	Month	TotaL_Sales
2014	1	14236.9
2014	2	4519.9
2014	3	55691.0
2014	4	28295.3
2014	5	23648.3
2014	6	34595.1
2014	7	33946.4
2014	8	27909.5
2014	9	81777.4
2014	10	31453.4
2014	11	78628.7
2014	12	69545.6
2015	1	18174.1
2015	2	11951.4
2015	3	38726.3
2015	4	34195.2
2015	5	30131.7
2015	6	24797.3
2015	7	28765.3
2015	8	36898.3
2015	9	64595.9
2015	10	31404.9
2015	11	75972.6
2015	12	74919.5
2016	1	18542.5
2016	2	22978.8
2016	3	51715.9
2016	4	38750.0
2016	5	56987.7
2016	6	40344.5

year	Month	TotaL_Sales
2016	7	39262.0
2016	8	31115.4
2016	9	73410.0
2016	10	59687.7
2016	11	79412.0
2016	12	96999.0
2017	1	43971.4
2017	2	20301.1
2017	3	58872.4
2017	4	36521.5
2017	5	44261.1
2017	6	52981.7
2017	7	45264.4
2017	8	63120.9
2017	9	87866.7
2017	10	77776.9
2017	11	118447.8
2017	12	83829.3

Sales By Segment

In [23]:

```
%%sql
select Segment,round(SUM(Sales),2) AS Total_Sales
From [Sample - Superstore]
Group by Segment
order by Total_Sales DESC;
```

```
 * mssql+pyodbc://DESKTOP-TANM3JG/Amazon?driver=ODBC+Driver+17+for+SQL+Server&truste
d_connection=yes
Done.
```

Out[23]:

Segment	Total_Sales
Consumer	1161401.34
Corporate	706146.37
Home Office	429653.15

In [ ]:

```
```